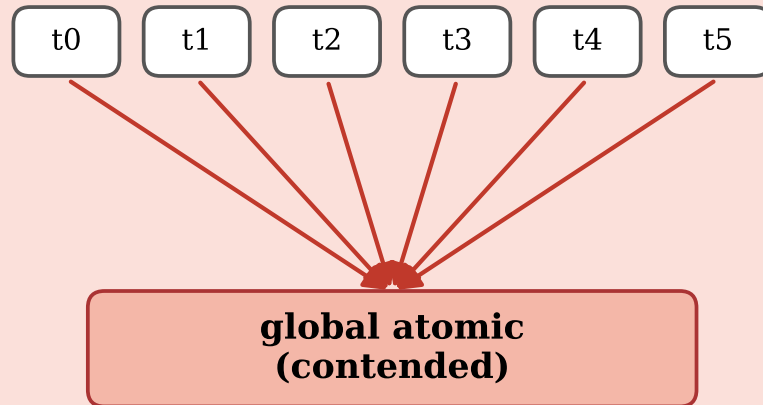


Where the reduction happens — and why it matters

mc_naive

one atomicAdd per path

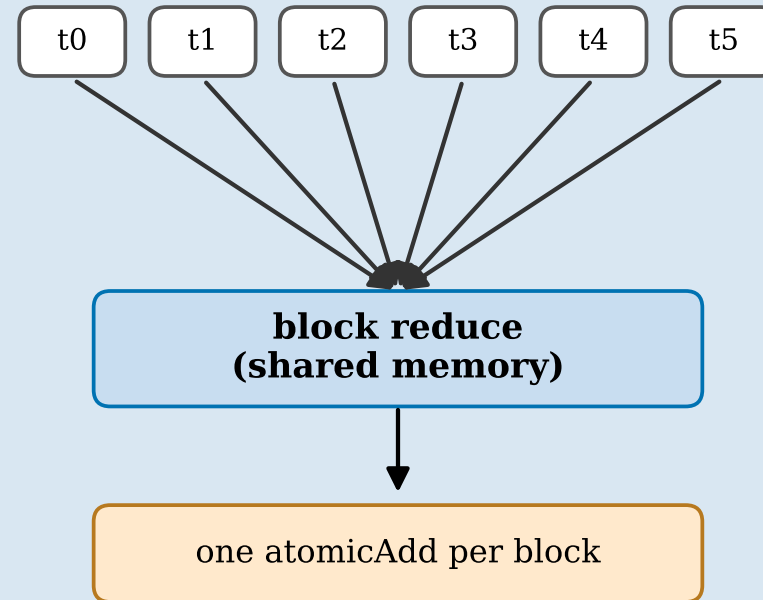


O(N) global atomics

compute SM = 2.8%
atomic contention

mc_shared

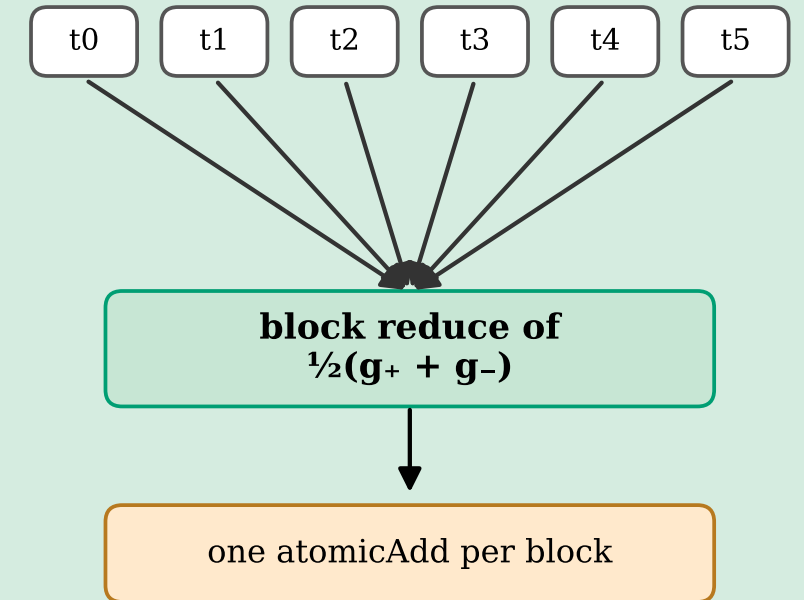
block-reduce + 1 atomic/block



compute SM = 79%
arithmetic-bound

mc_antithetic

+Z / -Z pair, then block-reduce



compute SM = 76%
 $\approx \frac{1}{2}$ variance